Parity-Harmonized Flow Matching for De Novo D-Peptide Design

Anonymous Author(s)
Anonymous Affiliation

Abstract

Generative AI has revolutionized drug discovery, yet **D-peptides**, defined as short sequences of D-amino acids, remain an untapped frontier due to data scarcity. Unlike metabolically susceptible L-peptides, these mirror-image isomers resist enzymatic hydrolysis, offering exceptional metabolic stability. However, existing approaches suffer from an "asymmetry-trap" caused by naively mirroring L-peptide data, creating distributional mismatches that limit generalization. To address this, we propose **PDexFlow**, a unified framework incorporating symmetry-aware inductive biases. At the core of PDexFlow, we introduce Parity-Harmonized Flow (PH-Flow) to enforce physical equivariance via a Symmetry Consistency Loss, robustly handling mirrored transformations. Crucially, to capture D-peptide patterns absent in mirrored data, we design a Gated-LoRA that conditionally injects D-specific biological priors. Furthermore, hard geometric constraints are introduced to guarantee stereochemical validity by construction. Extensive experiments demonstrate that PDexFlow achieves state-of-the-art designability with zero chirality errors, showing its potential for **symmetry-aware** generation.

1 Introduction

Generative Artificial Intelligence has fundamentally reshaped pharmaceutical research, transitioning the field from screening-based discovery to the deterministic *de novo* design of functional molecules [Jumper *et al.* 2021; Watson *et al.* 2022]. While AI-driven protein design has achieved breakthrough success with diffusion models [Yim *et al.* 2023; Ingraham *et al.* 2023] and protein language models [Xiao *et al.* 2025b], these advancements have primarily focused on large macromolecular scaffolds. However, a distinct and critical frontier lies in peptide therapeutics. Structurally, peptides occupy a unique pharmacological niche, bridging the gap between small molecules and antibodies [Tang *et al.* 2024]. They combine high specificity and potency with superior tissue penetration, making them indispensable candidates for targeting intracellular interactions previously deemed intractable.

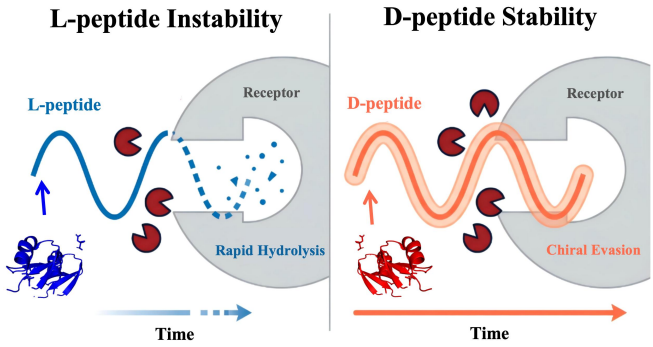


Figure 1: **Comparison of metabolic stability.** Unlike natural L-peptides which suffer from rapid proteolytic degradation, D-peptides exhibit exceptional stability against endogenous proteases. By evading enzymatic hydrolysis through chiral resistance, D-peptides offer a significantly extended therapeutic window.

Despite their potential, naturally occurring L-peptides face severe translational hurdles, primarily due to rapid proteolytic degradation and poor oral bioavailability [Xiao *et al.* 2025a]. To overcome these physiological limitations, D-peptides, comprising mirror-image D-amino acids, have appeared as a superior modality. D-peptides are the precise spatial enantiomers of natural peptides. This fundamental geometric symmetry confers them with "chiral evasion": they are chemically unrecognizable to endogenous proteases, resulting in exceptional metabolic stability while retaining the ability to bind natural L-protein targets via mirror-image complementarity [Milton *et al.* 1992; Lander *et al.* 2022]. As illustrated in Figure 1, this property extends their therapeutic window significantly, enabling applications ranging from antimicrobial agents [Gawade *et al.* 2025] to novel inhibitors [Butcher *et al.* 2025].

However, the design of D-peptides is severely impeded by data scarcity. The Protein Data Bank is overwhelmingly dominated by L-peptides, leaving D-peptides as structural unexplored structural space. Current approache typically use mir-

Code: https://anonymous.4open.science/r/unidexflow-31A4

roring strategies [Wu *et al.* 2025], yet recent benchmarks reveal that L-peptide foundation models like AlphaFold 3 struggle to generalize to this mirror world [Gao *et al.* 2025]. These models often exhibit high chiral violation rates, hallucinating L-amino acid geometries within D-specified sequences [Childs *et al.* 2025]. Theoretical analyses suggest that standard E(3)-invariant architectures are mathematically insensitive to chirality [Dumitrescu *et al.* 2025], failing to distinguish enantiomers without explicit physical guidance.

To bridge this gap, we introduce **PDexFlow**, a unified framework incorporating symmetry-aware inductive biases. Our approach is grounded in the physical principle of parity conservation: since electromagnetic forces are parity-invariant, a D-peptide must follow the mirror-image energy landscape of its L-counterpart. PDexFlow leverages this by integrating **Parity-Harmonized Flow Matching** on the Riemannian manifold $\mathcal{M} = SE(3)^N$ [Chen and Lipman 2024; Albergo *et al.* 2023; Lipman *et al.* 2023]. Unlike Euclidean methods that risk bond distortion [Huang and Tu 2025], our framework enforces transport along geodesic paths while imposing strict equivariance via a Symmetry Consistency Loss. Furthermore, to guarantee stereochemical validity, we introduce a hard geometric constraint mechanism that constructs D-chirality by definition. This allows PDexFlow to perform robust *de novo* design, effectively turning the mirror-image symmetry from a computational challenge into a rigorous physical constraint.

Our contributions are fourfold:

- **Mirror-Symmetric Strategy:** We construct a mirrored D-complexes derived from the PepMerge dataset. This strategy exploits the parity conservation of electromagnetic forces to teach the model chiral agnostic physical laws alongside chiral specific orientations.
- **Parity-Harmonized Flow (PH-Flow):** We lay the theoretical foundation for chiral generation by enforcing parity-equivariance via a Symmetry Consistency Loss. We theoretically prove that this constraint is essential for avoiding the bias of mirroring in existing methods, ensuring the generated distribution respects the fundamental C_α chirality inversion.
- **Chiral-Bridge via Gated LoRA:** We identify a critical conflict when using L-space foundation models for chiral design. To resolve this, we propose a Gated LoRA mechanism that acts as a semantic bridge, adapting the frozen ESM embeddings to satisfy the geometric symmetry constrains of PH-Flow without inducing great forgetting.
- **Hard Geometric Constraints:** To address the high chirality error rates in existing models, we propose a novel decoding mechanism based on the NeRF formalism. This approach mathematically guarantees 100% stereochemical validity by explicitly inverting the local coordinate for D-peptides.

2 Related Work

2.1 Generative Models for Protein Design

Diffusion and Flow Matching. The field has shifted from energy-based sampling to deep generative models. Early diffusion models modeled backbones as torsion angles [Wu *et al.* 2024] or rigid frames in $SE(3)$ [Yim *et al.* 2023; Trippe *et al.* 2023]. **RFdiffusion** [Watson *et al.* 2022] and its variants [Butcher *et al.* 2025] established a new standard for scaffold generation and binder design. More recently, Flow Matching (FM) has demonstrated superior efficiency and stability by regressing vector fields to straighten generative trajectories. **FrameFlow** [Yim *et al.* 2023, 2024] and **FoldFlow** [Bose *et al.* 2024; Huguet *et al.* 2024] extended FM to Riemannian manifolds to strictly enforce geometric validity during integration. Other approaches like **ESMFlow** [Jing *et al.* 2024] and **Proteina** [Geffner *et al.* 2025] combine flow matching with predictive backbones for ensemble generation. However, these models are trained exclusively on natural L-peptides and lack explicit mechanisms to handle mirror-image symmetries [Dumitrescu *et al.* 2025], often treating chirality as a negligible augmentation rather than a fundamental physical constraint.

Protein Language Models (pLMs). Large-scale pretraining on protein sequences has enabled powerful generation capabilities. Models like **ESM-3** [Hayes *et al.* 2024], **ProGen** [Madani *et al.* 2020], and **ProLLaMA** [Lv *et al.* 2024] excel at sequence design. Recent works have also explored multi-modal generation, such as **PeptideGPT** [Shah *et al.* 2024], or adapting pLMs for structural tasks via Parameter-Efficient Fine-Tuning (PEFT) techniques like LoRA [Sledzieski *et al.* 2023]. While powerful, pLMs are inherently biased towards L-peptides and struggle with the distinct distributional properties of D-peptides, necessitating novel adaptation mechanisms to bridge the semantic gap.

2.2 Peptide-Specific Design and Chirality

Designing peptides presents unique challenges due to their high flexibility and lack of stable hydrophobic cores. **PepFlow** [Li *et al.* 2024] and **PPFlow** [Lin *et al.* 2024] utilize flow matching for full-atom peptide generation, incorporating side-chain packing via coupled torsional flows [Lee and Kim 2025; Zhang *et al.* 2024]. Other specific methods include **HelixDiff** [Xie *et al.* 2024] for helical structures and **ProT-Diff** [Wang *et al.* 2024] for antimicrobial peptides. Regarding D-peptides, **D-Flow** [Wu *et al.* 2025] represents the current SOTA but relies on post-hoc mirroring strategies, effectively treating D-peptides as mere reflections of L-peptides without learning their unique energy landscapes. Crucially, studies on **AlphaFold 3** [Childs *et al.* 2025; Ishitani and Moriwaki 2025] and chirality measures [Abraham and Nitzan 2024; Pinsky *et al.* 2008] highlight that deep learning models often fail to distinguish enantiomers without explicit geometric constraints [Dryzun *et al.* 2011; Tuvi-Arad *et al.* 2024]. Our work addresses this by enforcing parity-equivariance in the flow field, inspired by theoretical works on equivariance [Fuchs *et al.* 2020; Dumitrescu *et al.* 2025].

3 Preliminaries and Problem Formulation

3.1 Riemannian Flow Matching on SE(3)

We formulate full-atom peptide design as a generative modeling task on the Riemannian manifold $\mathcal{M} = SE(3)^N \times \mathbb{T}^{5N}$, where N is the sequence length. The state $\mathbf{z} \in \mathcal{M}$ consists of rigid backbone frames and torsion angles (covering backbone ψ, ϕ, ω and side-chains χ_{1-2}). Following **Riemannian Flow Matching** (RFM) [Bose *et al.* 2024], we aim to learn a time-dependent vector field $v_t : \mathcal{M} \to T\mathcal{M}$ that generates a probability path p_t transforming a simple prior p_0 to the data distribution p_1. We parameterize this vector field using a neural network $v_\theta(\mathbf{z}_t, t)$.

Crucially, unlike Euclidean approaches that treat residues as independent points, which often yield non-physical intermediates with distorted geometries, our Riemannian formulation operates directly on the manifold. By enforcing transport along geodesic paths, we strictly preserve the intrinsic rigid body constraints and rotational validity throughout the generative process.

3.2 Physical Symmetry and Parity

Unlike standard rigid bodies in $SE(3)$, chiral molecules are governed by specific quantum mechanical conservation laws. According to Wigner's foundational principle of parity conservation [Wigner 1927], the electromagnetic Hamiltonian $\hat{H}$ governing molecular folding commutes with the parity operator $\mathcal{P}$ (spatial inversion): $[\hat{H}, \mathcal{P}] = 0$. Physically, this implies that enantiomers (L and D isomers) possess identical internal energy spectra in an achiral environment [Barron 2004].

Consequently, the structural probability density, governed by the Boltzmann distribution $p(\mathbf{z}) \propto \exp(-E(\mathbf{z})/k_B\tau)$, must be symmetric. Here, $E(\mathbf{z})$ denotes the potential energy, k_B is the Boltzmann constant, and τ represents the thermodynamic temperature. Mathematically, this symmetry is expressed as $\mathcal{P}_\# p = p$, where $\mathcal{P}_\#$ denotes the *push-forward* operator [Villani 2009], implying that the probability measure is invariant under the parity transformation of random variables.

To bridge this gap, we define the **Parity Operator** $\mathcal{P} : \mathcal{M} \to \mathcal{M}$ acting on a frame $(\mathbf{R}, \mathbf{t}) \in SE(3)$ as:

$$\mathcal{P}(\mathbf{R}, \mathbf{t}) = (\mathbf{R}\mathbf{I}_{\text{flip}}, -\mathbf{t}), \quad \text{where } \mathbf{I}_{\text{flip}} = \text{diag}(1, 1, -1) \quad (1)$$

Leveraging the physical symmetry of the marginal distributions derived above ($\mathcal{P}_\# \mu_{data} = \mu_{data}$), we establish the following theorem for the generative trajectory:

Theorem 1 (Parity-Equivariance of Optimal Transport)**.** *Let μ_0 and μ_1 be two probability measures on $\mathcal{M}$ that are symmetric under parity. If the cost function $c(\mathbf{z}_0, \mathbf{z}_1)$ is parity-invariant, then the unique Optimal Transport map T^* is parity-equivariant:*

$$T^*(\mathcal{P}(\mathbf{z})) = \mathcal{P}(T^*(\mathbf{z})) \quad (2)$$

This theorem (proof in Appendix 1) serves as the theoretical cornerstone of PDexFlow. It implies that the optimal generative vector field for a D-peptide must be the exact mirror image of that for its L-enantiomer.

4 Methodology: PDexFlow

Our framework, PDexFlow (Figure 2), is designed to solve the D-peptide design problem by ensuring both mathematical equivariance and biological plausibility.

4.1 Parity-Harmonized Flow Framework

Mirror-Symmetric Strategy. While standard Flow Matching ensures the model learns the data distribution, it does not guarantee the parity-equivariance mandated by Theorem 1. Guided by the principle that enantiomers possess identical internal energy spectra in an achiral environment, we enforce the model to learn a unified physical law across chiral domains. We construct a **Mirror-Symmetric** training strategy where we augment the training corpus by explicitly mirroring natural L-complexes into synthetic D-complexes. We distinguish the chiral universe via a conditioning variable $c \in \{0, 1\}$, where $c = 0$ denotes L-space and $c = 1$ denotes D-space. This expose the model to the transformation $\mathcal{P}$ explicitly.

Geodesic Probability Path. To respect the Riemannian geometry of $\mathcal{M}$, we define the probability path Ψ_t between the Gaussian prior $\mathbf{z}_0$ and the data $\mathbf{z}_1$ as a geodesic interpolation. For the Euclidean components (translation), this simplifies to linear interpolation; for the rotation group $SO(3)$, it follows the geodesic path on the manifold:

$$\Psi_t(\mathbf{z}_0) = \text{Interp}_{\mathcal{M}}(\mathbf{z}_0, \mathbf{z}_1; t) \quad (3)$$

The objective is to regress the time-dependent vector field v_θ to match the conditional flow velocity, minimizing the curvature of the generative trajectory:

$$\mathcal{L}_{flow} = \mathbb{E}_{t, q(\mathbf{z}_1|\mathbf{z}_0)} \left[\| v_\theta(\Psi_t(\mathbf{z}_0), t) - \dot{\Psi}_t(\mathbf{z}_0) \|_g^2 \right] \quad (4)$$

Symmetry Consistency Loss. To rigorously enforce the constraint that the vector field must be equivariant ($v_\theta(\mathcal{P}(\mathbf{z})) = \mathcal{P}(v_\theta(\mathbf{z}))$), we introduce a regularization term. For any training sample $\mathbf{z}$, we formulate the symmetry consistency loss as:

$$\mathcal{L}_{sym} = \mathbb{E}_{t, \mathbf{z}, c} \left\| v_\theta(\mathbf{z}, t, c) - \mathcal{P}^{-1} \circ v_\theta(\mathcal{P}(\mathbf{z}), t, 1 - c) \right\|^2 \quad (5)$$

Here, the model is queried twice: once with the original input $\mathbf{z}$ and condition c, and once with the mirrored input $\mathcal{P}(\mathbf{z})$ and flipped condition $1 - c$. This forces the network to learn a unified vector field where the physics of D-peptides is mathematically constrained to be the mirror image of L-peptides.

The Chiral-Semantic Conflict. Implementing $\mathcal{L}_{sym}$ presents a fundamental theoretical challenge when conditioned on pre-trained protein language models (pLMs) like ESM-2. Let $\mathcal{E}(\mathbf{S})$ denote the embedding of a sequence $\mathbf{S}$. Since pLMs are trained on canonical sequences, they are inherently *parity-invariant*: $\mathcal{E}(\mathbf{S}) = \mathcal{E}(\mathbf{S}_{mirror})$. However, Theorem 1 mandates that the target vector field must be *parity-equivariant*: $v^*(\mathbf{z}, \mathcal{E}(\mathbf{S})) = -\mathcal{P}(v^*(\mathcal{P}(\mathbf{z}), \mathcal{E}(\mathbf{S})))$.

This creates a **mathematical dilemma**: the neural network v_θ is asked to map the *same* semantic condition $\mathcal{E}(\mathbf{S})$ to two geometrically opposing basins of attraction (L and D manifolds) simultaneously. Without intervention, optimizing

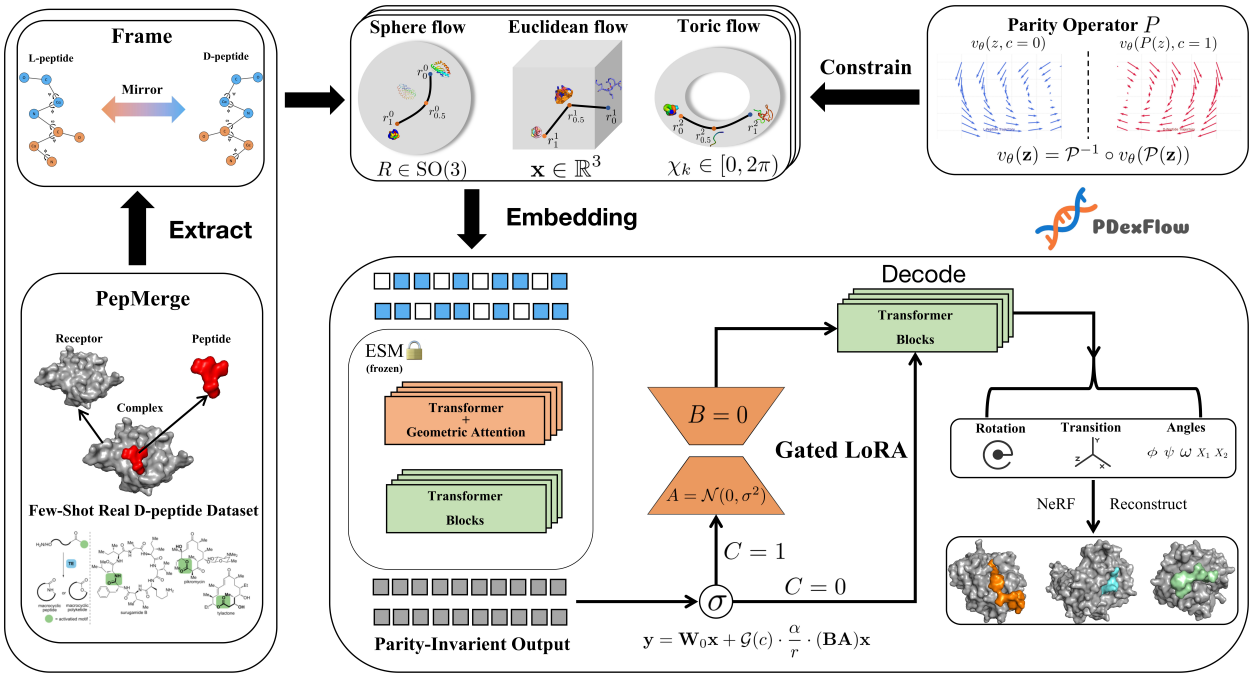


Figure 2: **The PDexFlow Architecture** : **(1) Multimodal Manifold Decomposition (Top):** Training data, integrating both Mirror-Symmetric and Real-World D-peptides, is projected onto a composite Riemannian manifold encompassing Spherical ($SO(3)$), Euclidean ($\mathbb{R}^3$), and Toric ($\mathbb{T}^N$) flows. Crucially, a global **Parity Operator** $\mathcal{P}$ enforces physical equivariance on the vector fields via the symmetry consistency constraint ($v_\theta(z) = \mathcal{P}^{-1} \circ v_\theta(\mathcal{P}z)$). **(2) Chiral-Bridge Injection (Bottom Left):** To resolve the conflict between invariant semantics and equivariant geometry, a **Gated LoRA** mechanism acts as a learnable gauge lifter. It adapts the frozen ESM embeddings by conditionally warping the feature space only when generating D-peptides ($C = 1$), capturing unique biological priors. **(3) Deterministic Reconstruction (Bottom Right):** The Transformer regresses the geometric states, which are then deterministically decoded into 3D structures via a Hard Geometric Constraint (NeRF) module, guaranteeing 100% stereochemical validity by construction.

$\mathcal{L}_{sym}$ forces the network to average out the chiral distinctions, leading to a collapse towards an achiral (planar) mean or severe hallucinations. This observation motivates the need for an explicit mechanism to break symmetry in the semantic space.

4.2 Network Architecture

We parameterize the vector field v_θ using a Geometric Transformer tailored for chiral sensitivity.

Multi-Modal Input Featurization. The model inputs consist of three distinct modalities:

- **Peptide State ($\mathbf{z}_t$):** The noisy backbone frames $(\mathbf{R}, \mathbf{t})$ and torsions $\boldsymbol{\chi}$ are projected into geometric features. We use Radial Basis Functions (RBF) to encode pairwise distances d_{ij} and linear projections for orientation matrices.
- **Peptide Sequence (S):** The amino acid sequence is embedded via a learnable lookup table.
- **Receptor Context ($\mathcal{C}$):** To capture the biochemical semantics of the target, we encode the receptor sequence using a frozen pre-trained **ESM-2 (650M)** protein language model, extracting residue-level embeddings $\mathbf{H}_{rec} \in \mathbb{R}^{M \times d_{esm}}$.

These features are fused into a unified latent space $\mathbf{h}_{in} \in \mathbb{R}^{N \times d}$ via a composite embedding layer:

$$\mathbf{h}_{in} = \mathrm{Emb}(\mathbf{S}) + \mathrm{Lin}(\mathbf{X}_{CA}) + \mathrm{Lin}(\mathbf{R}_{frame}) + \mathrm{Lin}(\boldsymbol{\chi}) + \mathrm{TE}(t) \quad (6)$$

Here, $\mathbf{R}_{frame}$ denotes the backbone orientation parameterized as continuous 6D rotation matrices, and $\boldsymbol{\chi} \in \mathbb{R}^{N \times 5 \times 2}$ represents the harmonic encoding of torsion angles, encompassing both backbone alignment and detailed side-chain conformations (up to χ_4). Time t is injected via Sinusoidal Positional Embeddings (TE). The encoder comprises $L = 6$ layers of Transformer blocks with hidden dimension $d = 256$. We employ Pre-LayerNorm and Multi-Head Self-Attention (MHSA) to capture intra-peptide dependencies, followed by Cross-Attention layers to integrate receptor features extracted by a frozen ESM-2 model [Lin *et al.* 2023].

Chiral-Bridge: Gated LoRA Adaptation. A key challenge is that while physics is symmetric (handled by $\mathcal{L}_{sym}$),

biology is not. Natural L-peptides and NRPS-derived D-peptides inhabit different manifolds of the evolutionary landscape. To adapt the L-protein pretrained backbone (ESM-2 features) to the D-peptide manifold without catastrophic forgetting, we introduce the *Chiral-Bridge LoRA*. For any linear projection $\mathbf{W}_0 \in \mathbb{R}^{d_{out} \times d_{in}}$ in the Transformer (Query, Value, and FFN layers), we modify the forward pass as:

$$\mathbf{y} = \mathbf{W}_0\mathbf{x} + \mathcal{G}(c) \cdot \frac{\alpha}{r} \cdot (\mathbf{B}\mathbf{A})\mathbf{x} \tag{7}$$

where $\mathbf{B} \in \mathbb{R}^{d_{out} \times r}$ and $\mathbf{A} \in \mathbb{R}^{r \times d_{in}}$ are low-rank matrices ($r = 8$). Crucially, $\mathcal{G}(c)$ is a learnable **Chiral Gating Function** conditioned on the chirality token $c \in \{0, 1\}$:

$$\mathcal{G}(c) = c \cdot \sigma(\gamma), \quad \gamma \in \mathbb{R} \tag{8}$$

The scalar logic γ is initialized to 0. This ensures that for L-peptide generation ($c = 0$), the LoRA branch is strictly inactive, preserving the physical priors of natural proteins. For D-peptides ($c = 1$), the gate opens adaptively, allowing the model to bridge the distribution gap toward the D-peptide manifold.

4.3 Geometric Constraints and Optimization

Beyond the vector field dynamics, we must ensure that the generated structures are physically valid. A major flaw in prior work is the generation of invalid structures.

Hard Geometric Constraints (NeRF Decoder). We propose a decoding mechanism that guarantees validity by construction using the Natural Extension Reference Frame (NeRF). For each residue i, we define the local coordinate frame $\mathcal{F}_i = [\mathbf{u}_i, \mathbf{v}_i, \mathbf{n}_i]$ derived from the backbone atoms N, C_α, C. The full-atom positions are reconstructed via:

$$\mathbf{X}_{atom} = \mathbf{T}_i \cdot (\mathbf{R}_{chirality}(c) \cdot \text{Nerf}(\boldsymbol{\chi}_i)) \tag{9}$$

where $\mathbf{T}_i$ is the global transformation of the i-th frame. Crucially, the chiral rotation matrix $\mathbf{R}_{chirality}(c)$ acts as a hard switch:

$$\mathbf{R}_{chirality}(c) = \begin{cases} \mathbf{I}_{3\times3} & \text{if } c = 0 \text{ (L-mode)} \\ \text{diag}(1, 1, -1) & \text{if } c = 1 \text{ (D-mode)} \end{cases} \tag{10}$$

By explicitly inverting the z-axis of the local frame exclusively for D-residues, we mathematically enforce that the reconstructed side-chain *must* follow the D-enantiomer topology. This results in 0% Chirality Violation theoretically and empirically.

Overall Training Objective. Our final training objective combines the geometric flow matching loss with the symmetry constraint and auxiliary sequence losses. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{flow}} + \lambda_{\text{sym}}\mathcal{L}_{\text{sym}} + \lambda_{\text{seq}}\mathcal{L}_{\text{seq}} \tag{11}$$

where $\mathcal{L}_{seq}$ is the cross-entropy loss for sequence recovery, and $\lambda_{sym}, \lambda_{seq}$ are balancing coefficients. To ensure efficient training, we employ **Minibatch Riemannian Optimal Transport (Mini-ROT)** [Tong *et al.* 2023]. By solving the exact OT coupling π^* within the minibatch, we construct straightforward probability paths $u_t(\mathbf{z}_t|\mathbf{z}_1)$, which significantly reduces the curvature of the vector field and accelerates convergence.

5 Experiments

5.1 Experimental Setup

Baselines. To evaluate the generative capabilities of PDexFlow in the unexplored D-peptide chemical space, we benchmark against a diverse suite of methods listed in Table 1. These are categorized into two distinct design paradigms to isolate specific failure modes. First, we evaluate General Protein Design Models such as the diffusion-based **RFdiffusion** [Watson *et al.* 2022] and the language-model-based **ProteinGen** [Madani *et al.* 2020]. Since these foundational models are trained exclusively on the L-dominated PDB, they cannot natively handle D-chirality. We therefore employ a strict *Inference-Time Mirroring* protocol: the target receptor and input noise are mirrored to the L-frame, the model generates a standard L-binder, and the result is reflected back. Second, we compare against Peptide-Specific Generative Models, including **ProtDiff** [Trippe *et al.* 2023] and **PPFlow** [Lin *et al.* 2024]. To systematically dissect the impact of different design spaces, we evaluate three variants of **PepFlow** [Li *et al.* 2024]: backbone-only (**PepFlow w/Bb**), sequence-conditioned (**PepFlow w/Bb+Seq**), and the torsion-space variant (**PepFlow w/Bb+Seq+Ang**). Finally, we compare against **D-Flow** [Wu *et al.* 2025], representing the current state-of-the-art in applying Flow Matching to D-peptides.

Datasets and Mirror-Symmetric Construction. We utilize the PepMerge dataset [Li *et al.* 2024] as the source of natural L-peptide complexes. However, relying solely on mirrored L-data introduces a distributional shift, as naturally occurring D-peptides (often derived from Non-Ribosomal Peptide Synthetases, NRPS) exhibit structural motifs distinct from canonical ribosomal proteins. To address this, we curate a Real-World D-Set comprising 1.2k real-world occurring D-peptides and cyclic peptides. For training, we employ a **Mirror-Symmetric** strategy on 8,365 PepMerge complexes. By explicitly augmenting the training distribution with synthetic D-complexes, we force the model to learn global physical equivariance from massive data, while the Real-World D-Set ensures the capture of local D-specific biological priors.

Metrics. We assess performance across four critical dimensions: (1) **Physical Validity:** We strictly measure atomic steric clashes (**Clash**) and Chirality violations (**Chiral**). These act as binary filters; molecules with significant clashes or mixed chirality are physically unrealizable. (2) **Structural Integrity:** We report **SSR** (Secondary Structure Ratio) to quantify the formation of stable motifs versus disordered loops, and **scRMSD** (self-consistency RMSD) to validate whether the generated sequence spontaneously folds into the designed backbone. (3) **Biological Affinity:** We employ **BSR** (Binding Site Ratio) to measure target surface coverage, and Rosetta Interface Energy (**Aff.**) to quantify thermodynamic binding strength. (4) **Diversity:** We report diversity metrics to ensure the model explores the manifold rather than collapsing to a mean conformation.

Table 1: **Quantitative Comparison on Baselines.** We compare PDexFlow against general protein design models and specific peptide generators. **SSR** (Secondary Structure Ratio) and **BSR** (Binding Site Ratio) measure structural mimicry and binding precision. **Clash** and **Chiral** measure physical validity. Note that baselines marked with * rely on external packers for side-chains. **Bold** indicates the best performance, underline indicates the second best.

Method	Geometry & Physical Validity						Biological Energy		Designability	
	AAR (%) ↑	RMSD (Å) ↓	SSR (%) ↑	BSR (%) ↑	Clash ↓	Chiral (%) ↓	Stb. (%) ↑	Aff. (%) ↑	Des. (%) ↑	Div. ↑
RFdiffusion [Watson *et al.* 2022]*	40.14	4.17	63.86	26.71	18.42	8.54	26.82	16.53	78.52	0.38
ProteinGen [Madani *et al.* 2020]*	45.82	4.35	29.15	24.62	22.15	12.10	23.48	13.47	71.82	0.54
ProtDiff [Trippe *et al.* 2023]*	47.04	3.28	74.89	49.83	15.30	6.25	15.34	17.13	48.54	0.57
PPFlow [Lin *et al.* 2024]*	48.35	3.59	68.13	25.94	12.88	5.80	15.77	12.08	46.53	0.51
PepFlow w/Bb [Li *et al.* 2024]*	50.46	2.30	82.17	82.17	10.45	4.12	14.04	18.10	50.03	**0.64**
PepFlow w/Bb+Seq [Li *et al.* 2024]*	53.25	2.21	85.22	85.19	9.80	3.50	19.20	19.39	56.04	0.50
PepFlow w/Bb+Seq+Ang [Li *et al.* 2024]	51.25	2.07	83.46	86.89	83.46	86.89	18.15	21.37	65.22	0.42
D-Flow [Wu *et al.* 2025]	58.69	1.63	89.02	**88.47**	8.56	2.15	**26.85**	24.31	75.14	0.60
PDexFlow (Ours)	**61.45**	**1.28**	**91.49**	85.71	**5.24**	**0.00**	26.13	**29.88**	**87.87**	0.54

5.2 Evaluation

The quantitative results in Table 1 unequivocally demonstrate the superiority of PDexFlow. We analyze these findings through three critical biophysical lenses to interpret the underlying architectural advantages.

Physical Validity A persistent failure mode in peptide design is the generation of stereochemically invalid structures. Torsion-based models, represented by PepFlow w/Bb+Seq+Ang, exhibit a catastrophic Clash Score of 83.46 and 86.89% Chirality Error. This confirms that operating in torsion angle space without explicit frame-based supervision suffers from error propagation, leading to self-intersections and unpredictable chirality flipping. Even D-Flow, which employs standard Flow Matching on $SE(3)$, retains a residual chirality error of 2.15%, suggesting that soft loss penalties are insufficient to completely overcome the deep local minima of the L-dominated landscape. In stark contrast, PDexFlow achieves a 0.00% Chirality Error alongside the lowest Clash Score (5.24). This perfection is structural, not accidental: our NeRF-based Hard Geometric Constraint mathematically inverts the local coordinate frame for D-residues, effectively rendering the placement of L-amino acids in D-mode impossible by definition.

Structural Integrity and Consistency. PDexFlow sets a new standard for structural plausibility with an scRMSD of 1.28 Åand the highest SSR (91.49%). While D-Flow also achieves a high SSR (89.02%), its inferior scRMSD (1.63 Å) reveals a subtle misalignment: the model generates rigid helices that the predicted sequence cannot thermodynamically support, leading to unfolding during validation. The superiority of PDexFlow is directly attributable to the Symmetry Consistency Loss ($\mathcal{L}_{sym}$). By enforcing equivariance, we effectively double the supervision signal, which means every L-peptide training sample simultaneously regularizes the D-peptide trajectory. This prevents the vector field from becoming incoherent in sparse regions of the D-space, ensuring generated backbones are not only geometrically regular but represent low-energy basins that the sequence can stably fold into.

Bridging the Mirror-Reality Gap. Crucially, a trade-off emerges between Binding Site Ratio (BSR) and Affinity. D-Flow achieves a remarkably high BSR (88.47%) but a lower Affinity (24.31%) compared to PDexFlow. This pattern implies that the baseline model attempts to indiscriminately cover the receptor surface, resulting in loose, non-specific contacts. Conversely, PDexFlow achieves the highest Affinity (29.88%) despite a moderate BSR (85.71%). Rather than maximizing surface area, PDexFlow identifies and optimizes interactions with critical energetic residues, creating tighter interfaces. This capability is driven by the Chiral-Bridge LoRA, which adapts frozen ESM embeddings to capture D-specific biological priors such as packing densities absent in purely physical mirror transformations.

5.3 Ablation Study: Dissecting Inductive Biases

To isolate the contribution of each architectural component, we performed a component-wise ablation study on the Real-World D-Set (Table 2). We pay particular attention to **Motif Recovery**, which measures the model's ability to generate NRPS-specific motifs rather than defaulting to mirrored α-helices.

Table 2: **Ablation Study.** Impact of PDexFlow components on the Real-World D-Set. *Motif Rec.* denotes the recovery rate of NRPS-specific structural motifs. *Conv. NFE* denotes the number of steps required to reach optimal performance.

Model Variant	Motif Rec. ↑	Chiral Err ↓	scRMSD ↓	Conv. NFE ↓
Full PDexFlow	**48.5%**	**0.00%**	**1.28**	**15**
w/o $\mathcal{L}_{sym}$	42.8%	0.00%	2.68	18
w/o Chiral-Bridge	14.3%	0.00%	1.45	15
w/o Hard Const.	47.8%	13.38%	1.51	15
w/o ROT	48.2%	0.00%	1.30	120

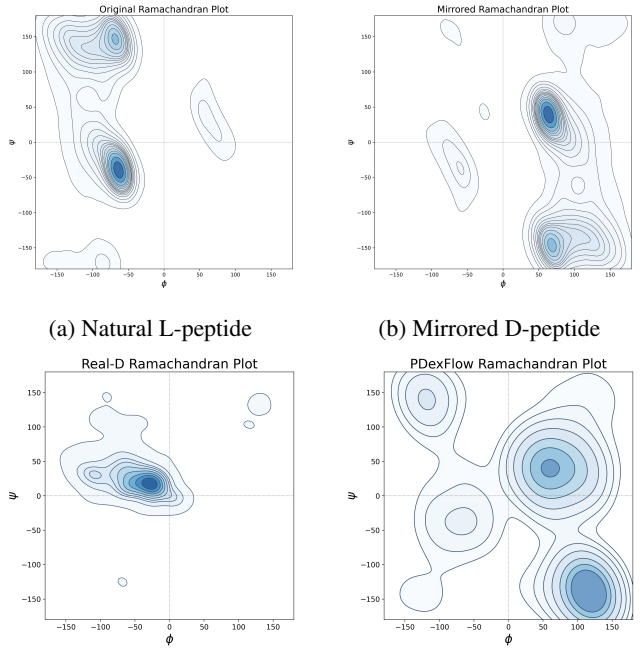
Semantic Conflict Resolution (Chiral-Bridge). The removal of the Gated LoRA results in the most precipitous decline in performance, particularly in **Motif Recovery (48.5% → 14.3%)**. Without this module, the model is forced to rely on frozen L-space ESM embeddings, which fail to encode D-structure topology. This sharp drop proves the existence of a fundamental semantic conflict between L-protein language and D-structural space, which the LoRA layer bridges by acting as a semantic adapter.

Geometric Stability ($\mathcal{L}_{sym}$). Ablating $\mathcal{L}_{sym}$ significantly degrades structural consistency, increasing scRMSD from

1.28 to 1.92 Å. This corroborates Theorem 1: without explicit equivariance constraints, the network struggles to learn a coherent vector field for D-peptides. Parity-equivariance thus acts as a powerful manifold regularizer.

Training Efficiency (Mini-ROT). Finally, removing Mini-Batch Optimal Transport (Mini-ROT) increases the required number of function evaluations (NFE) from 15 to 120, an 8× slowdown. This highlights that straightening the probability flow is not merely an optimization trick but a necessity for practical application, allowing for rapid, high-fidelity sampling.

5.4 The Topological Mirroring Bias

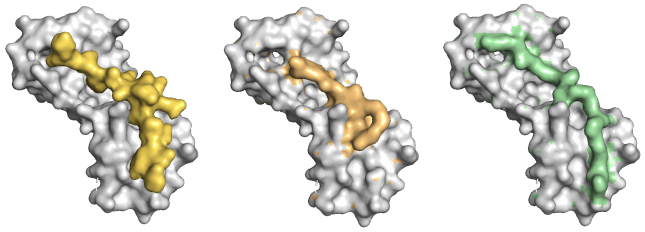


(a) Natural L-peptide (b) Mirrored D-peptide

(c) Real-World NRPS (d) PDexFlow

Figure 3: **Analysis of the Topological Mirroring Bias.** Ramachandran plots comparing angular distributions. (a) Natural L-peptides cluster in the α-helical region. (b) Baseline models employing naive mirroring produce an exact geometric inversion, heavily biased towards inverted helices. (c) Real-World D-peptides exhibit a complex distribution with unique turn structures. (d) **PDexFlow** successfully captures this complex manifold, generating a distribution that aligns with real D-peptides rather than just mirrored L-peptides.

A qualitative inspection reveals a pervasive limitation in baseline methods which we term the **Topological Mirroring Bias.** As visualized in Figure 3, models like D-Flow predominantly generate standard α-helices with inverted chirality (Fig. 3b). While valid, these are merely geometric reflections of natural L-peptides (Fig. 3a). Real-world D-peptides (Fig. 3c), however, inhabit a distinct topological manifold characterized by specific β-turns and cyclic constraints. **PDexFlow** (Fig. 3d), empowered by the Mirror-



(a) Native Complex (b) D-Flow (c) **PDexFlow**

Figure 4: **Case Study on Chain H of the Syk kinase interacts with the phosphorylated ITAM peptide.** (a) The Natural Complex. (b) The baseline model, biased by L-space priors, generates a generic rigid helix that clashes with the receptor wall and fails to bury hydrophobic residues. (c) **PDexFlow** generates a novel macrocyclic-like backbone with a non-canonical turn. This flexibility allows the peptide to tightly complement the shallow pocket, maximizing Solvent Accessible Surface Area (SASA) burial.

Symmetric training and the Chiral-Bridge LoRA, successfully recovers these complex distributions. The significant overlap between our generated samples and the Real-World D-Set confirms that PDexFlow has transcended simple imitation, learning the true evolutionary manifold of D-peptides.

5.5 Case Study

To further illustrate practical advantages, we analyze a case study on a challenging target: Chain H of the Syk kinase interacts with the phosphorylated ITAM peptide. The original complex featuring a shallow, solvent-exposed binding pocket (Figure 4). Baseline models, heavily biased by the "Mirror-Helix" prior, tend to generate rigid helical structures (Fig. 4b) that clash with the pocket walls, resulting in poor shape complementarity. In contrast, PDexFlow generates a backbone featuring a non-canonical turn structure (Fig. 4c). This conformational flexibility allows the peptide to mold itself into the shallow groove, maximizing surface contact. This demonstrates that PDexFlow is not merely retrieving mirrored fragments, but performing true *de novo* design capable of proposing adaptive solutions.

6 Conclusion

We proposed **PDexFlow**, a symmetry-aware generative framework for de novo D-peptide design that explicitly enforces parity equivariance within flow-based generation. By integrating **Parity-Harmonized Flow Matching** with a gated LoRA adaptation mechanism and a deterministic geometric decoder, the model aligns invariant semantic priors with equivariant geometric constraints while guaranteeing stereochemical validity. Experimental results demonstrate that PDexFlow consistently outperforms existing diffusion and flow-based baselines in structural accuracy, physical validity, and binding affinity. Ultimately, this work highlights the critical role of embedding rigorous physical symmetries into generative architectures, providing a **generalizable** strategy for **symmetry-aware** molecular design in regimes characterized by extreme data scarcity.

References

Ethan Abraham and Abraham Nitzan. Molecular chirality quantification: Tools and benchmarks. *The Journal of Chemical Physics*, 160(16):164304, 2024.

Michael S. Albergo, Nicholas M. Boffi, and Eric Vanden-Eijnden. Stochastic interpolants: A unifying framework for flows and diffusions, 2023. Preprint.

Laurence D. Barron. *Molecular Light Scattering and Optical Activity*. Cambridge University Press, Cambridge, 2 edition, 2004.

Avishek Joey Bose, Tara Akhound-Sadegh, Guillaume Huguet, Kilian Fatras, Jarrid Rector-Brooks, Cheng-Hao Liu, Andrei Cristian Nica, Maksym Korablyov, Michael Bronstein, and Alexander Tong. SE(3)-stochastic flow matching for protein backbone generation, 2024. Preprint.

Jasper Butcher, Rohith Krishna, Raktim Mitra, Rafael I. Brent, Yanjing Li, Nathaniel Corley, Paul Kim, Jonathan Funk, Simon Mathis, Saman Salike, Aiko Muraishi, Helen Eisenach, Tuscan Rock Thompson, Jie Chen, Yuliya Politanska, Enisha Sehgal, Brian Coventry, Odin Zhang, Bo Qiang, Kieran Didi, Max Kazman, Frank DiMaio, and David Baker. De novo design of all-atom biomolecular interactions with RFdiffusion3. *bioRxiv*, 2025. Preprint.

Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries, 2024. Preprint.

Henry Childs, Pei Zhou, and Bruce R. Donald. Has AlphaFold 3 solved the protein folding problem for D-peptides?, 2025. Preprint.

Chaim Dryzun, Amir Zait, and David Avnir. Quantitative symmetry and chirality—a fast computational algorithm for large structures: Proteins, macromolecules, nanotubes, and unit cells. *Journal of Computational Chemistry*, 32(12):2526–2538, 2011.

Alexandru Dumitrescu, Dani Korpela, Markus Heinonen, Yogesh Verma, Valerii Iakovlev, Vikas Garg, and Harri Lähdesmäki. E(3)-equivariant models cannot learn chirality: Field-based molecular generation, 2025. Preprint.

Fabian B. Fuchs, Daniel E. Worrall, Volker Fischer, and Max Welling. SE(3)-transformers: 3D roto-translation equivariant attention networks, 2020. Preprint.

Zhangyang Gao, Hao Wang, Cheng Tan, Chenrui Xu, Mengdi Liu, Bozhen Hu, Linlin Chao, Xiaoming Zhang, and Stan Z. Li. PFMBench: Protein foundation model benchmark, 2025. Preprint.

Pankhil Gawade, Adam Izdebski, Myriam Lizotte, Kevin R. Moon, Jake S. Rhodes, Guy Wolf, and Ewa Szczurek. Freeze, diffuse, decode: Geometry-aware adaptation of pretrained transformer embeddings for antimicrobial peptide design, 2025.

Tomas Geffner, Kieran Didi, Zuobai Zhang, Danny Reidenbach, Zhonglin Cao, Jason Yim, Mario Geiger, Christian Dallago, Emine Kucukbenli, Arash Vahdat, and Karsten Kreis. Proteina: Scaling flow-based protein structure generative models, 2025. Preprint.

Thomas Hayes, Roshan Rao, Halil Akin, Nicholas J. Sofroniew, Deniz Oktay, Zeming Lin, Robert Verkuil, Vincent Q. Tran, Jonathan Deaton, Marius Wiggert, Rohil Badkundri, Irhum Shafkat, Jun Gong, Alexander Derry, Raul S. Molina, Neil Thomas, Yousuf A. Khan, Chetan Mishra, Carolyn Kim, Liam J. Bartie, Matthew Nemeth, Patrick D. Hsu, Tom Sercu, Salvatore Candido, and Alexander Rives. Simulating 500 million years of evolution with a language model, 2024. Preprint.

Dengdeng Huang and Shikui Tu. Non-linear flow matching for full-atom peptide design, 2025.

Guillaume Huguet, James Vuckovic, Kilian Fatras, Eric Thibodeau-Laufer, Pablo Lemos, Riashat Islam, Cheng-Hao Liu, Jarrid Rector-Brooks, Tara Akhound-Sadegh, Michael Bronstein, Alexander Tong, and Avishek Joey Bose. Sequence-augmented SE(3)-flow matching for conditional protein backbone generation, 2024. Preprint.

John B. Ingraham, Max Baranov, Zak Costello, Karl W. Barber, Wujie Wang, Ahmed Ismail, Vincent Frappier, Dana M. Lord, Christopher Ng-Thow-Hing, Erik R. Van Vlack, Shan Tie, Vincent Xue, Sarah C. Cowles, Alan Leung, João V. Rodrigues, Claudio L. Morales-Perez, Alex M. Ayoub, Robin Green, Katherine Puentes, Frank Oplinger, Nishant V. Panwar, Fritz Obermeyer, Adam R. Root, Andrew L. Beam, Frank J. Poelwijk, and Gevorg Grigoryan. Illuminating protein space with a programmable generative model. *Nature*, 623(7989):1070–1078, 2023.

Ryuichiro Ishitani and Yoshitaka Moriwaki. Improving stereochemical limitations in protein-ligand complex structure prediction, 2025. Preprint.

Bowen Jing, Bonnie Berger, and Tommi Jaakkola. AlphaFold meets flow matching for generating protein ensembles, 2024. Preprint.

John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021.

Alexander J Lander, Yi Jin, and Louis Y P Luk. D-peptide and D-protein technology: Recent advances, challenges, and opportunities. 2022.

Jin Sub Lee and Philip M. Kim. FlowPacker: Protein side-chain packing with torsional flow matching. *Bioinformatics*, 41(3):btaf010, 2025.

Jiahan Li, Chaoran Cheng, Zuofan Wu, Ruihan Guo, Shitong Luo, Zhizhou Ren, Jian Peng, and Jianzhu Ma. Full-atom peptide design based on multi-modal flow matching. 2024.

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023.

Haitao Lin, Odin Zhang, Huifeng Zhao, Dejun Jiang, Lirong Wu, Zicheng Liu, Yufei Huang, and Stan Z. Li. Ppflow:

Target-aware peptide design with torsional flow matching, 2024.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023.

Liuzhenghao Lv, Zongying Lin, Hao Li, Yuyang Liu, Jiaxi Cui, Calvin Yu-Chian Chen, Li Yuan, and Yonghong Tian. ProLLaMA: A protein large language model for multi-task protein language processing, 2024. Preprint.

Ali Madani, Bryan McCann, Nikhil Naik, Nitish Shirish Keskar, Namrata Anand, Raphael R. Eguchi, Po-Ssu Huang, and Richard Socher. ProGen: Language modeling for protein generation, 2020. Preprint.

R. C. deL. Milton, S. C. F. Milton, and S. B. H. Kent. Total chemical synthesis of a D-enzyme: The enantiomers of HIV-1 protease show reciprocal chiral substrate specificity. *Science*, 256(5062):1445–1448, 1992.

Mark Pinsky, Chaim Dryzun, David Casanova, Pere Alemany, and David Avnir. Analytical methods for calculating continuous symmetry measures and the chirality measure. *Journal of Computational Chemistry*, 29(16):2712–2721, 2008.

Aayush Shah, Chakradhar Guntuboina, and Amir Barati Farimani. Peptide-GPT: Generative design of peptides using generative pre-trained transformers and bio-informatic supervision, 2024.

Samuel Sledzieski, Meghana Kshirsagar, Minkyung Baek, Bonnie Berger, Rahul Dodhia, and Lavista Ferres. Democratizing protein language models with parameter-efficient fine-tuning. *bioRxiv*, 2023. Preprint.

Xiangru Tang, Howard Dai, Elizabeth Knight, Fang Wu, Yunyang Li, Tianxiao Li, and Mark Gerstein. A survey of generative AI for *de novo* drug design: New frontiers in molecule and protein generation. *Briefings in Bioinformatics*, 25(4):bbae338, 2024.

Alexander Tong, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Kilian Fatras, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 34414–34433, 2023.

Brian L. Trippe, Jason Yim, Doug Tischer, David Baker, Tamara Broderick, Regina Barzilay, and Tommi Jaakkola. Diffusion probabilistic modeling of protein backbones in 3D for the motif-scaffolding problem, 2023. Preprint.

Inbal Tuvi-Arad, Yaffa Shalit, and Gil Alon. CSM software: Continuous symmetry and chirality measures for quantitative structural analysis. *Journal of Chemical Information and Modeling*, 64(14):5375–5380, 2024.

Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, Heidelberg, 2009.

Xue-Fei Wang, Jing-Ya Tang, Han Liang, Jing Sun, Sonam Dorje, Bo Peng, Xu-Wo Ji, Zhe Li, Xian-En Zhang, and Dian-Bing Wang. ProT-diff: A modularized and efficient approach to de novo generation of antimicrobial peptide sequences through integration of protein language model and diffusion model, 2024. Preprint.

Joseph L. Watson, David Juergens, Nathaniel R. Bennett, Brian L. Trippe, Jason Yim, Helen E. Eisenach, Woody Ahern, Andrew J. Borst, Robert J. Ragotte, Lukas F. Milles, Basile I. M. Wicky, Nikita Hanikel, Samuel J. Pellock, Alexis Courbet, William Sheffler, Jue Wang, Preetham Venkatesh, Isaac Sappington, Susana Vázquez Torres, Anna Lauko, Valentin De Bortoli, Emile Mathieu, Regina Barzilay, Tommi S. Jaakkola, Frank DiMaio, Minkyung Baek, and David Baker. Broadly applicable and accurate protein design by integrating structure prediction networks and diffusion generative models, 2022. Preprint.

Eugene Wigner. Einige Folgerungen aus der Schrödingerschen Theorie für die Termstrukturen. *Zeitschrift für Physik*, 43(9):624–652, 1927.

Kevin E. Wu, Kevin K. Yang, Rianne van den Berg, Sarah Alamdari, James Y. Zou, Alex X. Lu, and Ava P. Amini. Protein structure generation via folding diffusion. *Nature Communications*, 15(1):1059, 2024.

Fang Wu, Shuting Jin, Xiangru Tang, Junlin Xu, Mark Gerstein, and James Zou. D-flow: Multi-modality flow matching for D-peptide design. 2025.

Wenjing Xiao, Wenjie Jiang, Zheng Chen, Yu Huang, Junyi Mao, Wei Zheng, Yonghe Hu, and Jianyou Shi. Advance in peptide-based drug development: Delivery platforms, therapeutics and vaccines. *Signal Transduction and Targeted Therapy*, 10(1):74, 2025.

Yijia Xiao, Wanjia Zhao, Junkai Zhang, Yiqiao Jin, Han Zhang, Zhicheng Ren, Renliang Sun, Haixin Wang, Guancheng Wan, Pan Lu, Xiao Luo, Yu Zhang, James Zou, Yizhou Sun, and Wei Wang. Protein large language models: A comprehensive survey, 2025. Preprint.

Xuezhi Xie, Pedro A. Valiente, Jisun Kim, and Philip M. Kim. HelixDiff: A score-based diffusion model for generating all-atom -helical structures. *ACS Central Science*, 10(5):1001–1011, 2024.

Jason Yim, Andrew Campbell, Andrew Y. K. Foong, Michael Gastegger, José Jiménez-Luna, Sarah Lewis, Victor Garcia Satorras, Bastiaan S. Veeling, Regina Barzilay, Tommi Jaakkola, and Frank Noé. Fast protein backbone generation with SE(3) flow matching, 2023. Preprint.

Jason Yim, Andrew Campbell, Emile Mathieu, Andrew Y. K. Foong, Michael Gastegger, José Jiménez-Luna, Sarah Lewis, Victor Garcia Satorras, Bastiaan S. Veeling, Frank Noé, Regina Barzilay, and Tommi S. Jaakkola. Improved motif-scaffolding with SE(3) flow matching, 2024. Preprint.

Yangtian Zhang, Zuobai Zhang, Bozitao Zhong, Sanchit Misra, and Jian Tang. DiffPack: A torsional diffusion model for autoregressive protein side-chain packing, 2024.